

Tennis Match Data Analysis

Data Analysis Bootcamp Project

September 2026

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Introduction

Tennis Match Data Analysis

Data Analysis Bootcamp Project — project README

Overview

A data analysis bootcamp project: cleaning and analysing a raw tennis match dataset, then answering a set of analytical questions about players, matches, and betting and fan behavior.

Team

Member	GitHub	Questions
Yasaman Aghaei	@YasamanAghaei	Q3, Q4, Q5, Q6, Q9, Q11, Q14 + 2 optional
Mehrnoush Yaghoubi	@mehratty	Q1, Q2, Q10, Q13, Q15 + 2 optional
Keyvan Hajizadeh	@keyvanhj	Q7, Q8, Q12, Q16, Q17 + 2 optional

About the dataset

The dataset used in this project is private and is not included in this repository.

All data files (*.parquet, *.csv) are excluded through .gitignore. Because the data is private, and out of respect for the database creator's privacy and at their explicit request, the dataset will not be made available here.

The dataset consists of 15 parquet tables linked by match_id, covering match results, player information, set scores, tournament and venue details, point-by-point data, betting odds, and fan votes.

structure

```
tennis-data-analysis/
|  `-- tennis_dataset_schema.pdf      # full schema documentation
|-- cleaning/
|   |-- clean_tennis_data_en.py      # builds the 15 tables defined in the schema doc
|   |-- cleaning_7.ipynb             # statistics, pbp, power, odds, venue, votes, season
|   |-- cleaning_8.ipynb             # event, home_team, away_team, home_team_score,
|   |                                # away_team_score, time, tournament, round
|   |-- final_join.py                # builds matches_clean.csv / players_clean.csv
|   `-- Tennis_Scoring_Rule_Validator.ipynb
|
|                                # flags matches with impossible set scores
|                                # (tennis-rule check)
|-- keyvan_hajizadeh/                # Q7, Q8, Q12, Q16, Q17 + 2 optional
|-- yasaman_ghaei/                  # Q3, Q4, Q5, Q6, Q9, Q11, Q14 + 2 optional
`-- mehrnoush_yaghoubi/              # Q1, Q2, Q10, Q13, Q15 + 2 optional
```

Questions answered so far

Part 1 — required questions

1. How many tennis players are included in the dataset?
2. What is the average height of the players?
3. Which player has the highest number of wins?
4. What is the longest match recorded in terms of duration?
5. How many sets are typically played in a tennis match?
6. Which country has produced the most successful tennis players?
7. What is the average number of aces per match?
8. Is there a difference in the number of double faults based on gender?
9. Which player has won the most tournaments in a single month?
10. Is there a correlation between a player's height and their ranking?

11. What is the average duration of matches?
12. What is the average number of games per set in men's matches compared to women's matches?
13. What is the distribution of left-handed versus right-handed players?
14. What is the most common type of surface used in tournaments?
15. How many distinct countries are represented in the dataset?
16. Which player has the highest winning percentage against top-10 ranked opponents?
17. What is the average number of breaks of serve per match?

Part 2 — bonus / self-designed questions

1. Does the winner usually hit more aces?
2. Does the court surface affect the number of aces?
3. Does match duration depend on the court surface?
4. Do players' performances change with the court surface (e.g. clay vs. hard court)?
5. Is the fan voting share systematically biased toward the home player?
6. Comparing public/fan opinion with the betting market: which one predicted better?

Known data-quality findings

Missing player rows: roughly 27–31% of matches have no player information rows (`home_team` / `away_team`). This is a structural gap in the source data, not a cleaning error.

Incomplete round data: the round table covers only about 55% of matches.

Ranking snapshot: `current_rank` reflects a player's ranking at the time the data was collected, not their ranking at the time of each specific match.

Impossible set scores: about 2.2% of matches contain an impossible set score (fewer than 6 total games in a set). The investigation traced these to genuine retirements, concentrated at the lower tournament tiers (ITF / Challenger), rather than a data-collection bug.

For the full investigation, see: `docs/tennis_dataset_schema.pdf` and `cleaning/Tennis_Scoring_Rule_Validator.ipynb`.

Tennis scoring-rule validator (data-quality layer)

What it does

Aligns sets_played between home and away score tables; classifies every set under correct tennis rules (standard 6–x / 7–5 / 7–6, match/super TB, rare advantage sets); flags genuine anomalies (e.g. illegal 7–x scores), tiebreak-field mismatches, winner-vs-games contradictions, best-of-3 overruns, implausible durations, and completed matches with no set data.

Key outputs written back

- Aligned sets_played on both score tables
- Event flags: score_ready, set_scores_valid, is_retirement_like
- Full flag table: tennis_data_quality_flags.csv

Scale of issues found (from the notebook run)

- ~1,490 matches with at least one bad set status
- ~1,721 set-level anom_7 cases
- 564 tiebreak-field inconsistencies
- 64 winner-vs-games contradictions (many legitimate retirements)
- 0 best-of-3 overruns once only valid completed sets are counted
- 17 implausible durations outside 15–300 min
- 51 completed matches with no set data (walkover-like)
- Final usable counts: score_ready ≈ 14,413; set_scores_valid ≈ 14,822; is_retirement_like ≈ 275

Rules used as the baseline

- A standard completed set is 6 games (at least) with a margin of at least 2, or 7–5, or 7–6 after a set tiebreak.
- Scores of the form 7–0 through 7–4 are invalid (anom_7).
- A match / super tiebreak (10–x, win by 2) is allowed only on the deciding set.
- An incomplete *last* set is treated as a retirement / default pattern; an incomplete set that is *not* last is a genuine anomaly.
- In best-of-3, after a 2–0 in sets there must be no third set.
- The retired player is the loser; winner_code is never rewritten to “fix” the games count.

Data problems addressed

1. **Disagreement on sets_played between home and away** — recompute from non-null period_* columns and take the max of the two sides so both score tables agree.
2. **Impossible 7–x set scores** — flagged; feeds set_scores_valid.
3. **Rows with no player entities and no set periods** (tie-level / non-match grain) — excluded from set-scoring checks.
4. **Hidden retirements** — marked with is_retirement_like when the match is completed and has a winner but the winner has fewer than two sets by a naive games count.
5. **Incomplete tiebreak fields and walkovers** — diagnostic flags only; scores are not invented.
6. **Implausible duration** — flagged separately; the primary duration screen for analysis remains duration_reliable (plus the 15–300 minute band where used).

How the question notebooks use it

status	condition	meaning
valid_normal	6–0 ... 6–4, or 7–5	Standard completed set
valid_tiebreak	7–6	Set decided by a set tiebreak
valid_super_tb	≥10 games with margin ≥2, and the set is the deciding set	Match / super tiebreak
valid_advantage	8–6, 9–7, ...	Advantage set without a set tiebreak (special formats)
anom_7	7–0 through 7–4	Impossible under standard scoring rules
incomplete_last_set	Does not reach a valid endpoint and is the last set played	Retirement / default pattern
genuine_anomaly	Does not reach a valid endpoint and the match continued past it	True scoring inconsistency
anomaly_one_sided_null	Only one side has a score recorded	Incomplete recording

Question 1

How many unique tennis players are in the dataset?

Answer:

2,644 unique players (from 24,079 total home/away appearances across 16,873 matches). On average, each player appears in about 9.1 matches.

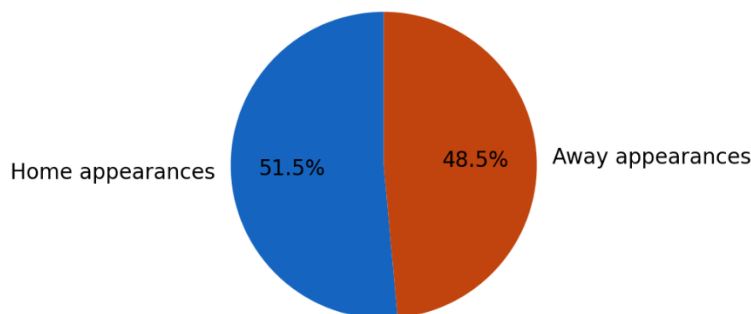
Side investigation:

is the "home"/"away" label meaningful for players, or just a random display convention?

Step 1 -- Row-level check (Chart 1):

Appearances split almost evenly: 51.5% home vs 48.5% away. No bias at the raw-data level.

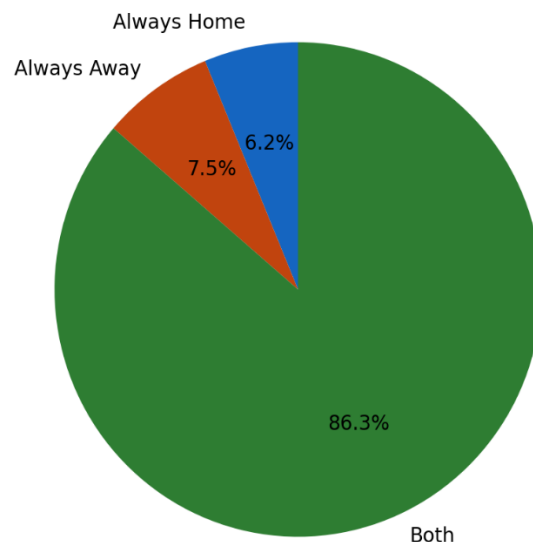
Chart 1: Share of appearances -- Home vs Away
(counted per match-row, not per player)



Step 2 -- Player-level check (Chart 2):

Of the 2,644 unique players, 86.3% (2,282) appeared on both sides at some point, 7.5% (197) were always "away", and 6.2% (165) were always "home". At first glance, the 13.7% who were always on one side looked like a possible hidden rule in how the source data assigns home/away.

Chart 2: Unique players -- Always Home / Always Away / Both
(counted per player, not per row)



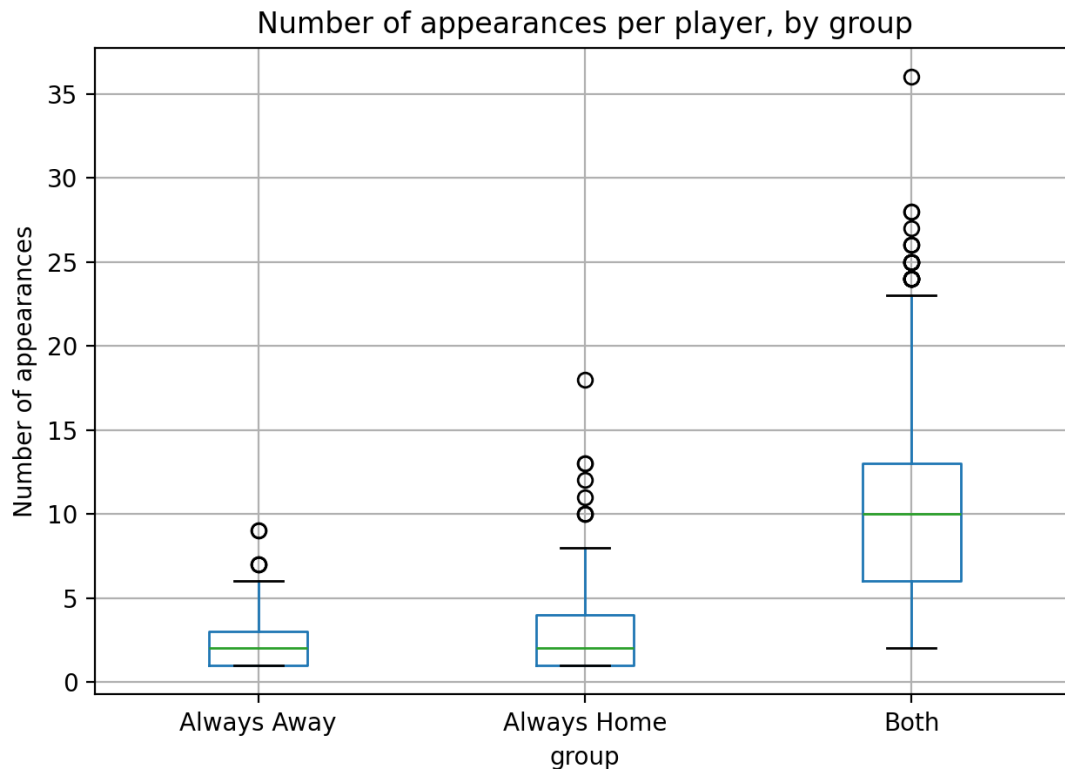
Step 3 -- Statistical follow-up:

Grouped players by number of appearances. The median player in both "Always Home" and "Always Away" groups had only 2 total appearances, versus a median of 10 for players who appeared on both sides. This means most of the 13.7% are explained by ordinary chance: a player with only 1-2 matches has a high probability of landing on the same side every time even under purely random assignment.

However, 52 players had 5 or more appearances while still always being on one side -- far too many to be explained by chance (e.g. a player with 18 appearances, always "home", has under a 0.001% probability of that happening randomly).

Conclusion:

The home/away label is effectively random for the vast majority of players, BUT a small subgroup (52 of 2,644 players, ~2%) shows a statistically significant, systematic pattern of always being assigned to one side. This is likely a rule in the source data's labeling convention (e.g. based on ranking or seeding at the time), not a data quality error.



Question 2

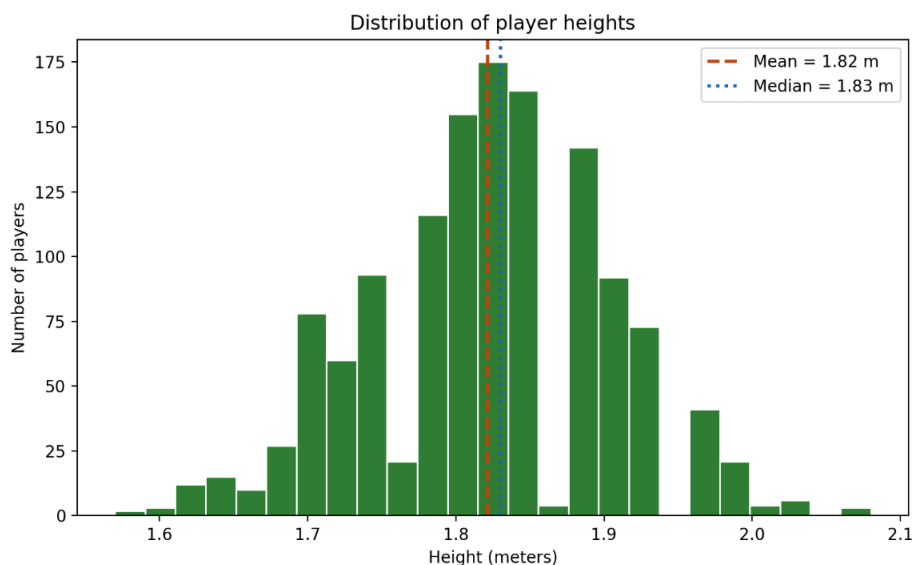
What is the average height of the players?

Results:

group	mean	median	std	count
Women	1.732 m	1.73 m	0.061	n = 256
Men	1.843 m	1.85 m	0.069	n = 1,060
Overall (mixed)	1.82 m	-	-	-

Interpretation:

There is a clear ~11 cm gap between the average heights of men and women in this dataset, consistent with known real-world averages in professional tennis (WTA ~1.70-1.75 m, ATP ~1.85 m).



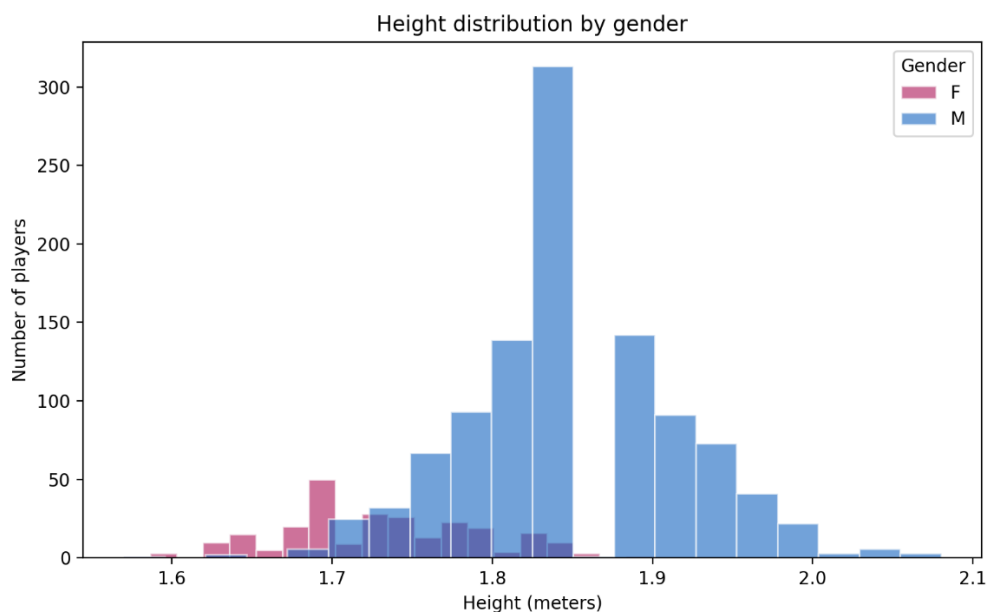
The overlaid histogram confirms the distribution is genuinely bimodal:

a smaller cluster of women around 1.65-1.75 m and a larger cluster of men around 1.78-1.95 m, with modest overlap in between.

This explains why the earlier single combined histogram looked broader/less clean than a typical single bell curve -- it was actually a mixture of two separate populations.

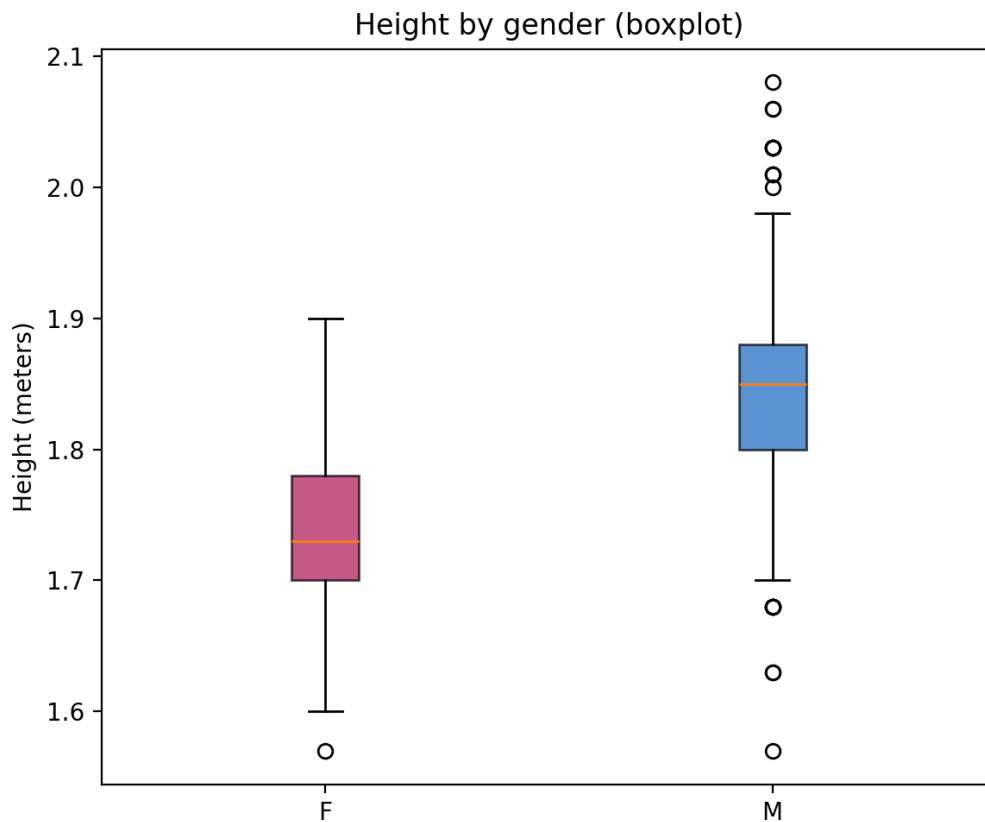
The overall mixed average (1.82 m) sits much closer to the men's mean than the women's, simply because men (1,060) outnumber women (256) roughly 4-to-1 in this height-available subset -- NOT because men's height "matters more". Reporting a single overall average without noting this imbalance would be misleading.

Standard deviations are similar between genders (0.061 vs 0.069), meaning the within-group spread is comparable -- the main difference is a shift in the mean, not a difference in variability.



Side observation:

the boxplot shows several low outliers in the men's group (around 1.57-1.68 m), unusually short for professional male players, though this was not confirmed as an error.



Question 3

Which player has the highest number of wins?

Finding

Popko leads with 28 wins, followed by Chidekh (22) and Jianu / Gengel (21 each). Ranking rests on 11,674 wins that could be linked to a player row — about 81% of completed matches that have a recorded winner. The remaining 2,739 wins were dropped because the winning side had no player row in home_team / away_team.

Method

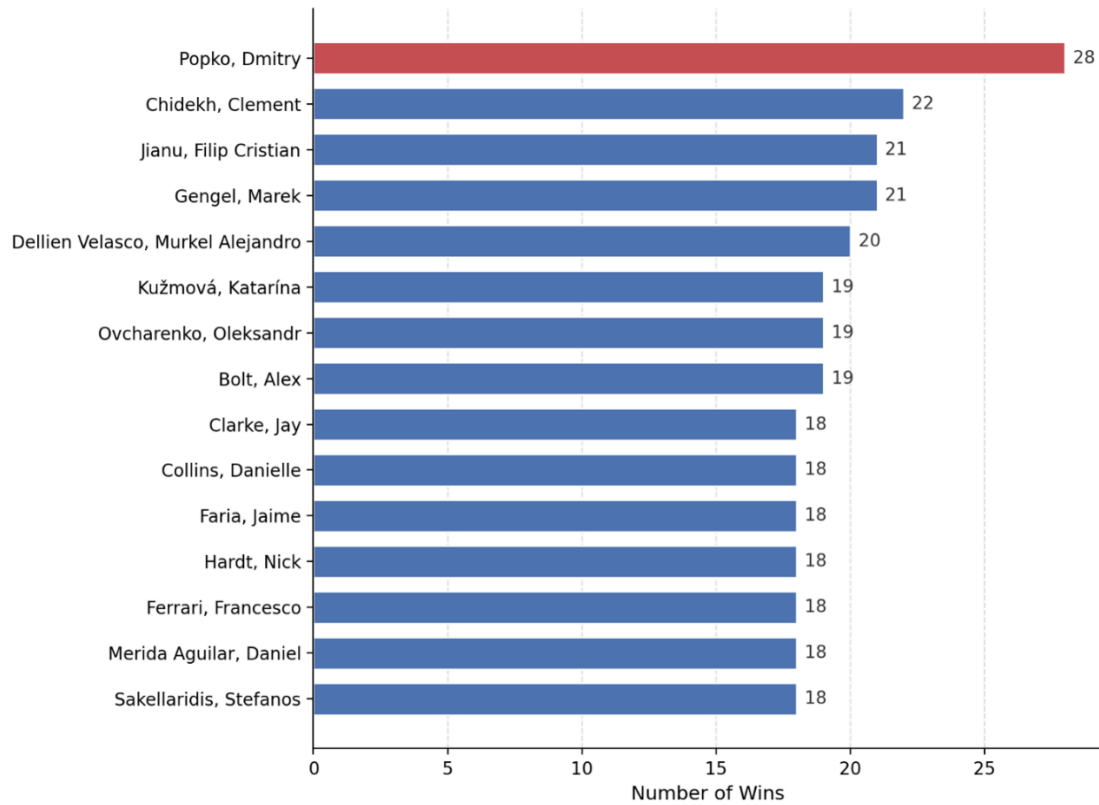
Completed matches with a non-null winner_code and score_ready; wins attributed via winner_code (1 = home, 2 = away); group by player_id / full_name.

Meaning

Total wins mix two things: how often a player competed and how often they won. ITF-heavy schedules can produce higher totals than fewer, harder matches at higher tiers. The ranking is a volume ranking, not a quality ranking.

Coverage is incomplete by construction. Without a minimum-match filter or tier split, the list overweights players who simply played more

Top 15 Players by Total Match Wins



Question 4

What is the longest match recorded in terms of duration?

Finding

After filters, the longest match is Simakin vs Bertrand (Monastir, ITF Men, hard outdoor), 4h 57m over 35 games (2024-02-10). The top 10 sit between roughly 4.4h and 5.0h.

Method

Start from duration_reliable and duration in **15–300 minutes**; join score tables; compute seconds per game; keep rows with positive total games. Same reliability pipeline used later in Q11 and surface-duration work.

Meaning

The upper bound of the candidate pool is partly an artifact of the 300-minute cap. The result is “longest inside the range treated as plausible,” not a claim that no real match exceeded five hours.

Duration window (15–300 minutes)

The duration analyses (Q4 longest match, Q11 average duration, and surface-duration comparisons) restrict to matches whose recorded length falls between **15 and 300 minutes**. That window is not arbitrary; it follows from the cleaned time table and from known upper bounds of completed professional matches.

Lower bound (15 minutes)

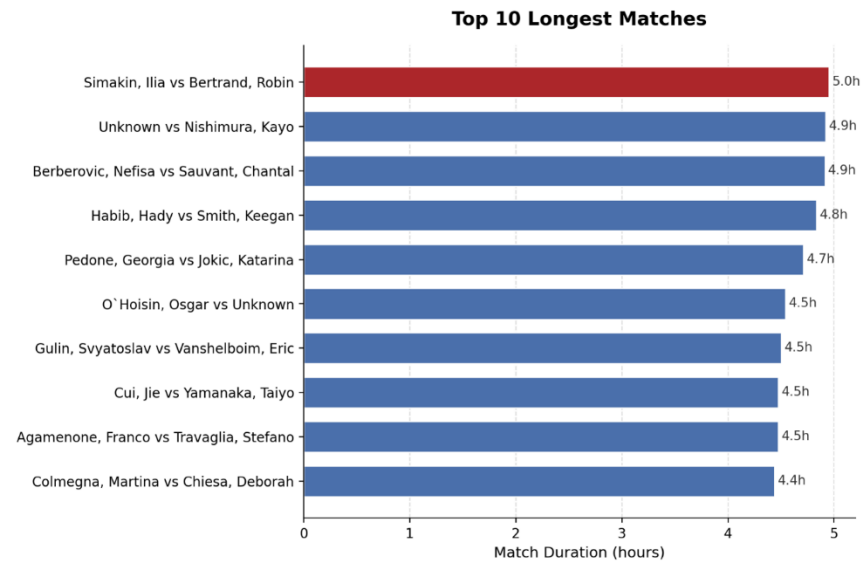
In the cleaned time data, only **3** rows (about **0.018%** of the table) have duration under 15 minutes. All of them have `duration_reliable = False`. They correspond to matches that did not run to a normal finish (retirement-like / incomplete recording), not to full best-of-3 contests. Excluding them removes noise without discarding a material share of completed matches.

Upper bound (300 minutes)

Only **14** rows (about **0.083%** of the table) exceed 300 minutes. Even when `duration_reliable` is True for some of those rows, values above five hours sit outside the range supported by completed professional matches in this dataset and by established records. Keeping them would let a handful of extreme (and often suspect) clock values dominate “longest match” rankings and inflate means.

Domain anchors for the upper tail

- Longest Olympic men’s singles match: Federer vs del Potro, London 2012 Olympic semi-final — **4h 26m** (ITF / Guinness).
- Long 3-set match with a third-set tiebreak: Skatov vs Tseng, ATP Challenger (clay, 2022) — about **4h 27m** (ATP archives).
- Long 3-set ATP Masters on clay: Nadal vs Djokovic, Madrid 2009 semi-final — about **4h 3m** (ATP).



Question 5

How many sets are typically played in a tennis match?

Finding

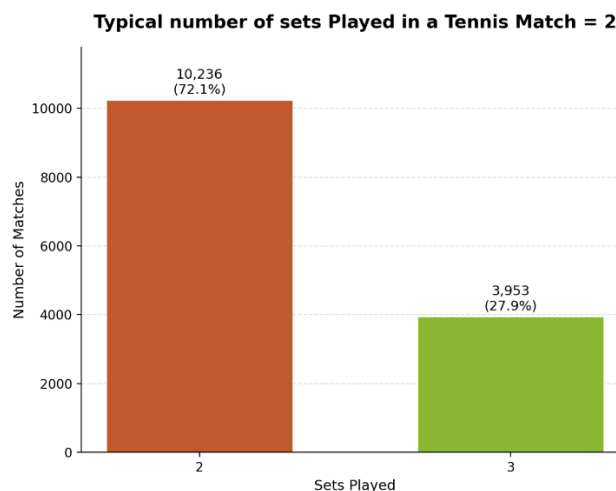
Of 14,189 completed matches: **72.1% end in 2 sets, 27.9% in 3** (mean 2.28). Agreement with the independent power table is 100% on 10,074 overlapping matches.

Method

Align sets_played from home and away score tables (max of the two); restrict to match_completed; cross-check max set_num from power.

Meaning

Matches the best-of-3 structure: straight-set wins stop at two sets, so a ~72/28 split means most matches do not reach a decider.



Question 6

Which country has produced the most successful tennis players?

Finding

By total wins: France (986), Italy (933), USA (909).

With ≥ 50 players, wins per player: Argentina (5.58), Brazil (5.00), France (4.88), Spain (4.87), Czech Republic (4.83).

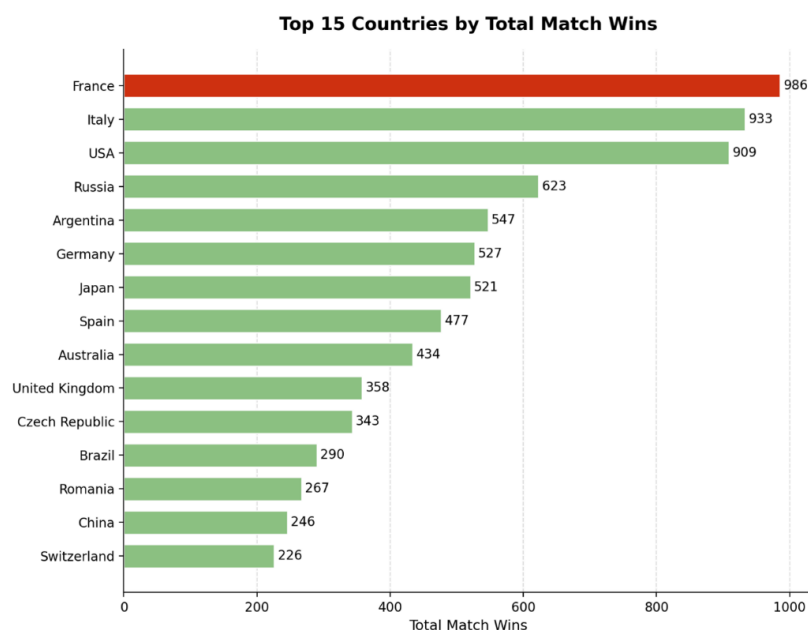
Method

Same win attribution as Q3, then group by country

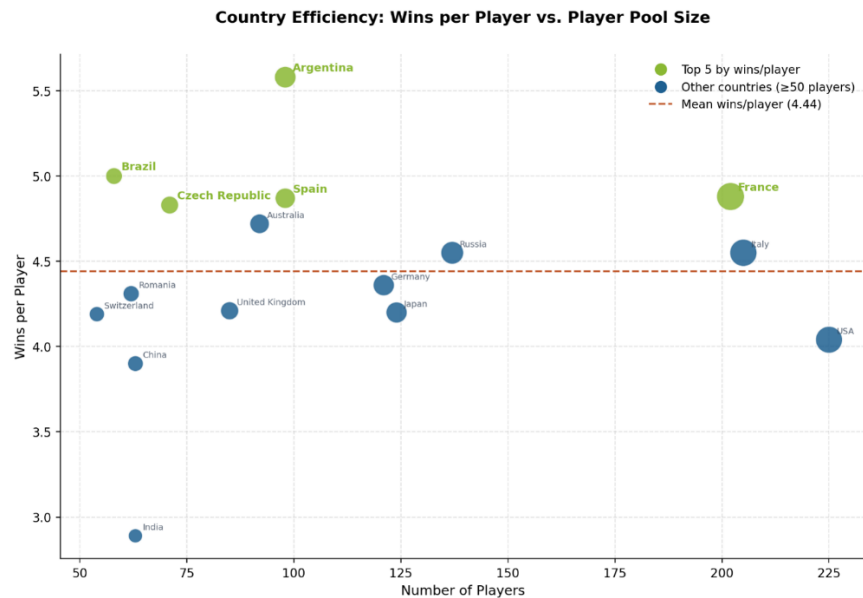
secondary metric = total wins / distinct players with a minimum player-count threshold.

Meaning

“Most successful” is not a single fact. Total wins reward large player pools; wins per player reward efficiency but is sensitive to the cutoff. At 50 players the two rankings overlap more than at a low threshold.



Answer depends on definition and on where the player-count bar is set. A few players with multiple country labels exist and were left as diagnostic noise rather than force-merged.



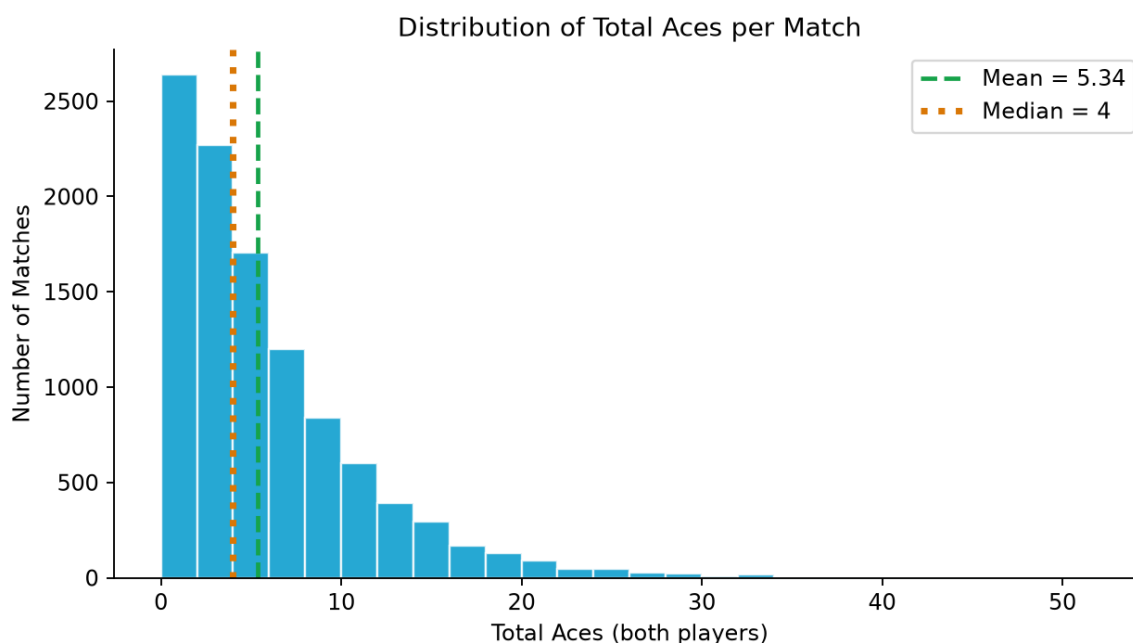
Question 7

What is the average number of aces per match?

The mean total number of aces per match is 5.34, while the median is 4; this discrepancy indicates a right-skewed distribution, which is evident in the chart.

Additionally, approximately 50% of the matches feature between 0 and 6 aces, suggesting that most matches are not serve-dominated.

Some outliers appear on the right side of the chart; these are genuine data points—stemming from long, five-set matches—and were therefore retained, causing the distribution to skew to the right.



Question 8

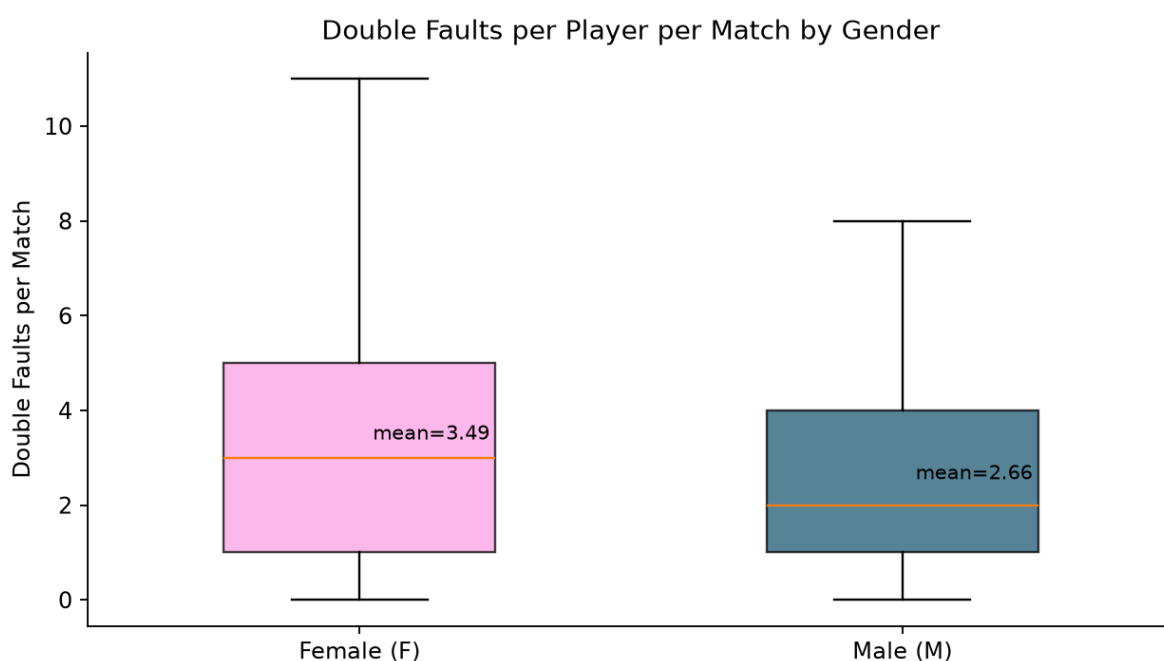
Is there a difference in the number of double faults based on gender?

The mean number of double faults for female players is 3.49, which is noticeably higher than the 2.66 recorded for men; the median values—3 versus 2—reflect this same pattern.

Observing the pink box plot (representing women), one can see a larger interquartile range, indicating greater dispersion in the number of double faults among female players.

Given the large sample size, the results are reliable; thus, it can be concluded that women commit significantly more double faults than men.

(Outliers have been excluded from the plot to improve readability.)



Question 9

Which player has won the most tournaments in a single month?

Finding

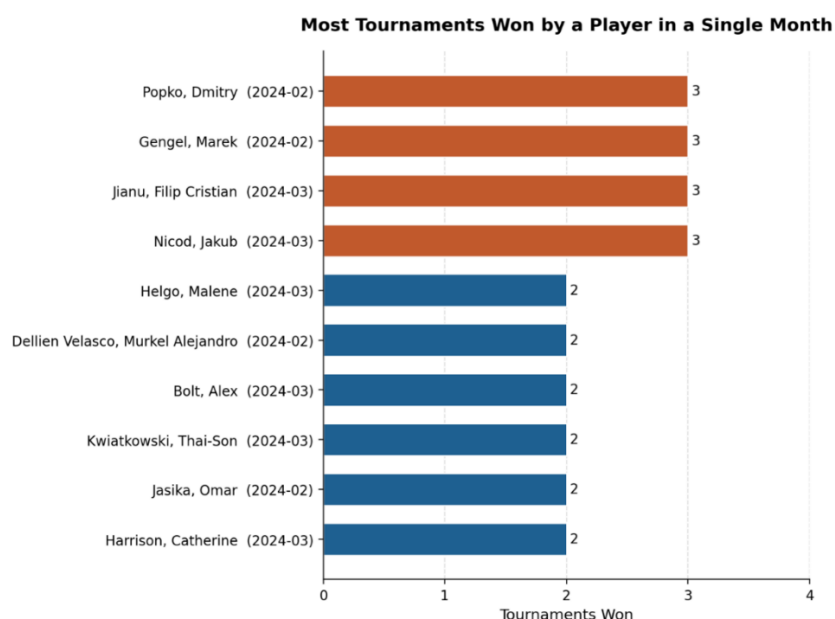
Maximum is **3 titles in one month**, reached by four players (Popko, Gengel, Jianu, Nicod) in Feb–Mar 2024, all ITF Men. Of 260 main-draw finals, 28 had no recorded winner and were excluded.

Method

Restrict to main draw (exclude qualifying); take `is_final`; require completed match with winner; attribute player as in Q3; group by player × month.

Meaning

Three events in one month fits weekly ITF calendars. The same pattern is less likely at ATP level because events are spaced further apart and fields are deeper. Measured only over months present in the data, not a multi-year season.



Question 10

Is there a correlation between a player's height and their ranking?

Answer:

Pearson $r = 0.106$, based on 1,307 players (49.4% of the 2,644 unique players -- the rest are missing either height or current_rank).

Interpretation:

The correlation is positive but very weak ($|r| < 0.2$ is generally considered negligible). This means there is NO meaningful linear relationship between a player's height and their current ranking in this dataset. The scatter plot confirms this visually: at every height value, players are spread across the full range of rankings (from top to bottom), rather than clustering toward better rankings for taller players.

The slightly positive sign (taller $\rightarrow$ numerically higher/worse rank) runs opposite to the common belief that height is an advantage in tennis (e.g. for serving) -- but given the near-zero magnitude, this is not a meaningful effect and could easily be noise.

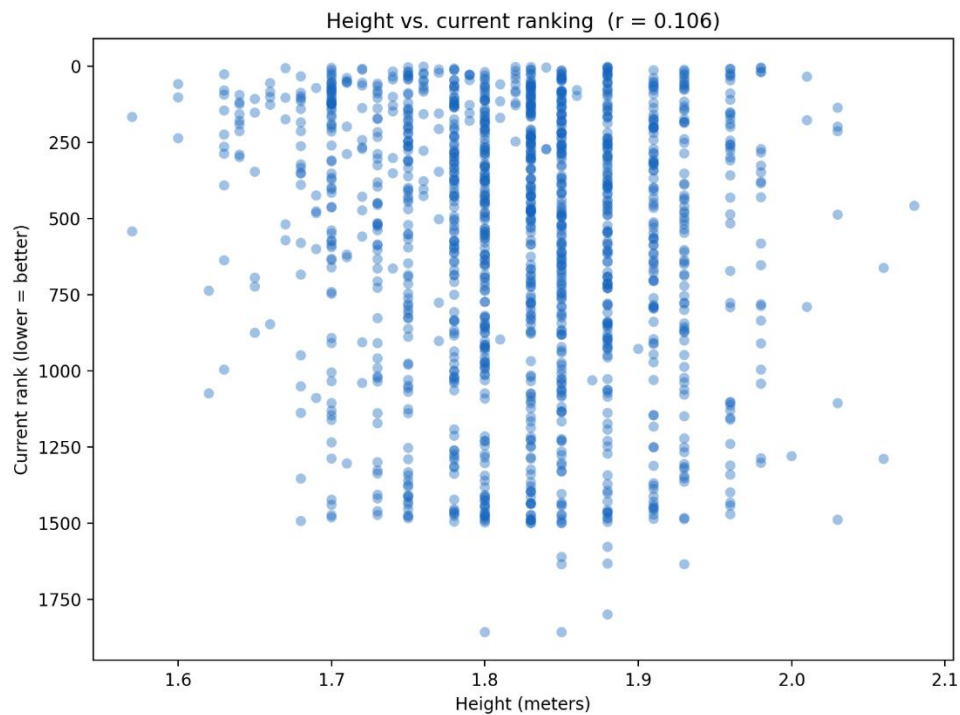
Side observation:

heights in the scatter plot form distinct vertical bands (e.g. 1.70, 1.73, 1.75, 1.78 m) rather than a smooth continuous spread. This suggests heights were originally recorded in another unit (likely feet/inches)

and converted to rounded meter values -- a data characteristic worth noting, not an error.

Caveat:

this analysis only covers players with BOTH height and current_rank recorded (49.4% of all players); it should not be generalized to the full player population without that context.



Question 11

What is the average duration of matches?

Finding

Same filters as Q4: mean **103.6 min**, median **96.8**, sd **35.2** (n = 10,075).

By sets: ~87.8 min (2 sets) vs ~139.6 min (3 sets).

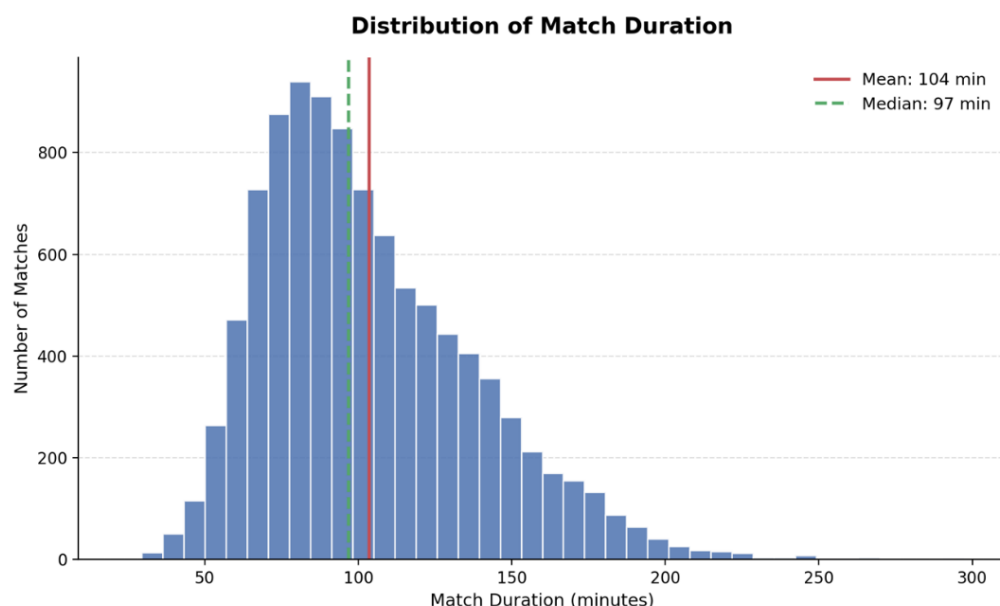
Method

duration_reliable, 15–300 minutes, completed matches, games cross-check.

Meaning

Mean above median indicates right skew. The ~52-minute gap between 2- and 3-set matches is consistent with the cost of one extra full set.

This analysis Describes the filtered population only. Retirements and rows with corrupted or incomplete duration are outside scope by design.



Question 12

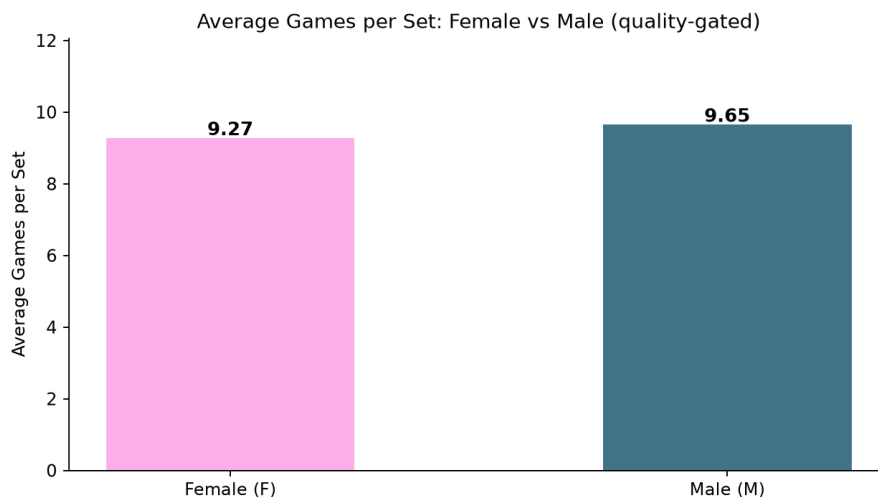
What is the average number of games per set in men's matches compared to women's matches?

The average number of games per set is 9.65 for men and 9.27 for women, indicating a difference of approximately half a game.

Observing the median—which is 9 for both groups—reveals that the discrepancy lies in longer sets and tie-breaks.

This slight difference may stem from the fact that women play best-of-three sets while men play best-of-five, or it could result from the differing scoring structures in men's five-set matches or the greater competitiveness of men's matches, rather than reflecting a difference in performance.

(Prior to calculation, 617 incomplete sets—such as 0-0 or 3-2 scores resulting from retirements or walkovers rather than a natural conclusion—were identified and excluded from the total of 20,461 sets to prevent the average from being artificially skewed downward.)



Question 13

What is the distribution of left handed versus right-handed players?

Answer (among players with a recorded value):

Right-handed: 1,013 players (88.4%)

Left-handed: 133 players (11.6%)

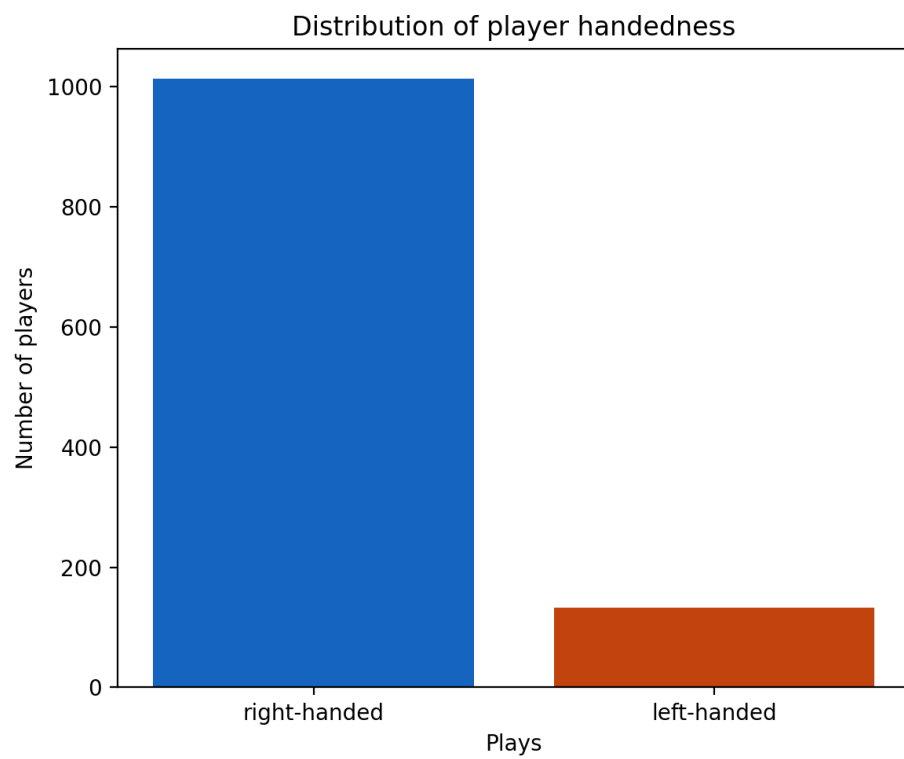
Interpretation:

The 11.6% left-handed rate is consistent with known real-world patterns: roughly 10% of the general population is left-handed, and professional tennis is commonly observed to have a slightly higher share of left-handed players than the general population (a left handed player's shots/spin are less familiar to right-handed-dominant opponents, a modest tactical edge). This result is unsurprising and does not indicate a data quality issue.

Coverage caveat:

This distribution is based on only 1,146 of 2,644 unique players (43.3%) -- the remaining 1,498 players (56.7%) have no recorded 'plays' value.

The result should be reported as "among players with known handedness", not generalized to the full player population.



Question 14

What is the most common type of surface used in tournaments?

Finding

Hard is first at both levels (~62% of matches, ~61% of tournaments); Clay second (~35–36%).

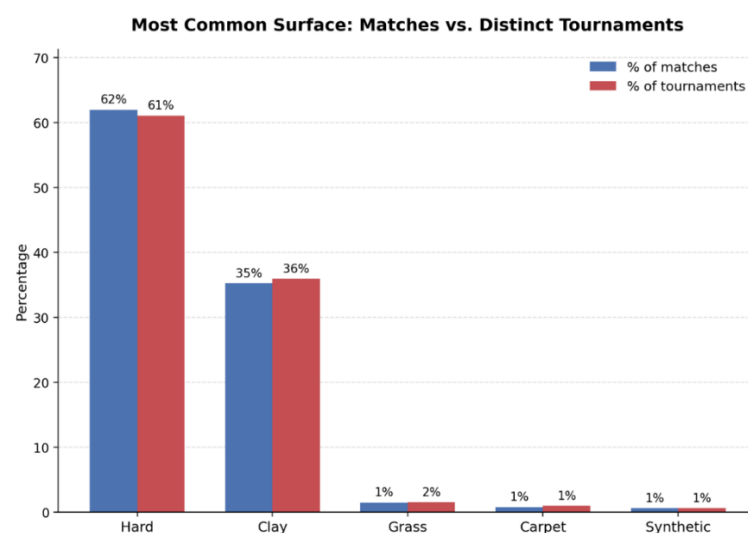
Method

Match-level counts from `ground_type`; tournament-level via mode of surface within (`tournament_name`, category). Broad families: Hard / Clay / Grass / Carpet / Synthetic.

Meaning

Match-level and tournament-level shares are close, so draw-size differences by surface are small in this dataset.

8 of 529 tournament groups had mixed surface labels (mode used). Ranking is unchanged; the inconsistency is real data noise, not silently fixed.

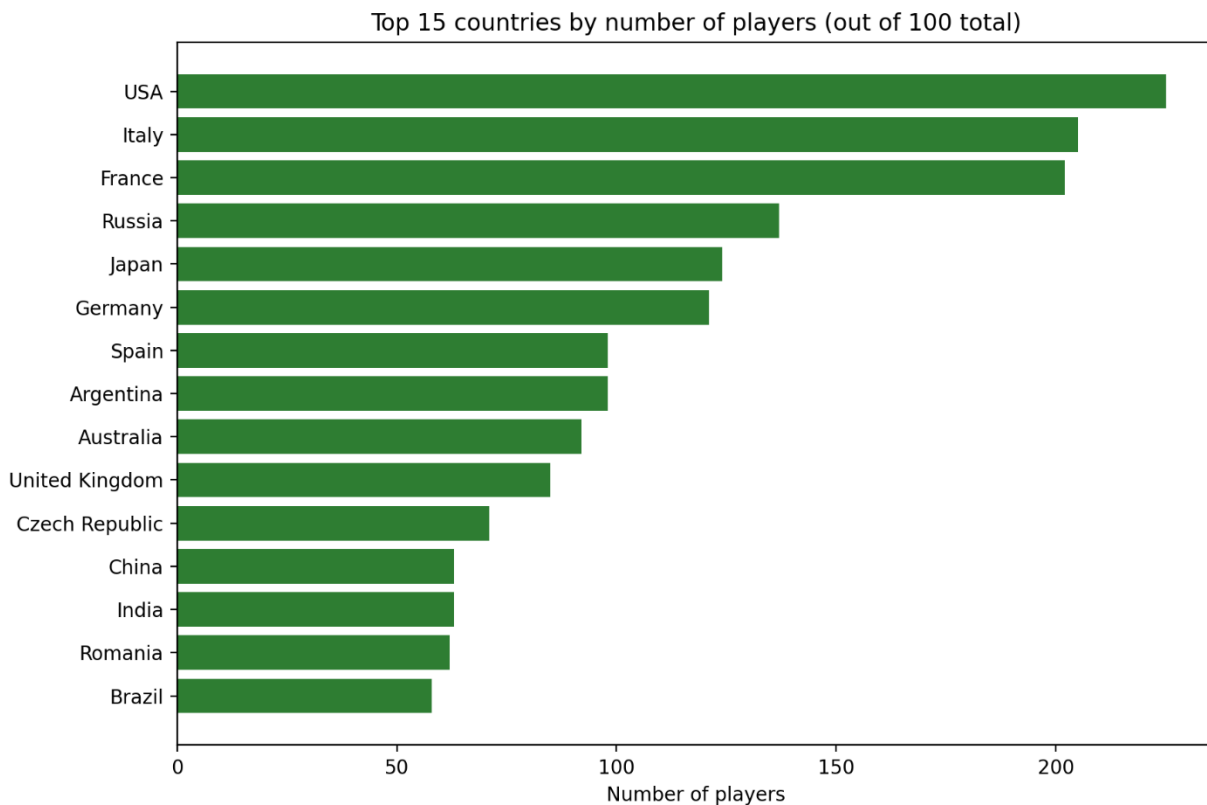


Question 15

How many distinct countries are represented in the dataset?

Answer

Distinct countries represented: 100



Goal:

the chart above is for a quick visual read; this full table is for completeness/appendix purposes, e.g. to paste into a report.

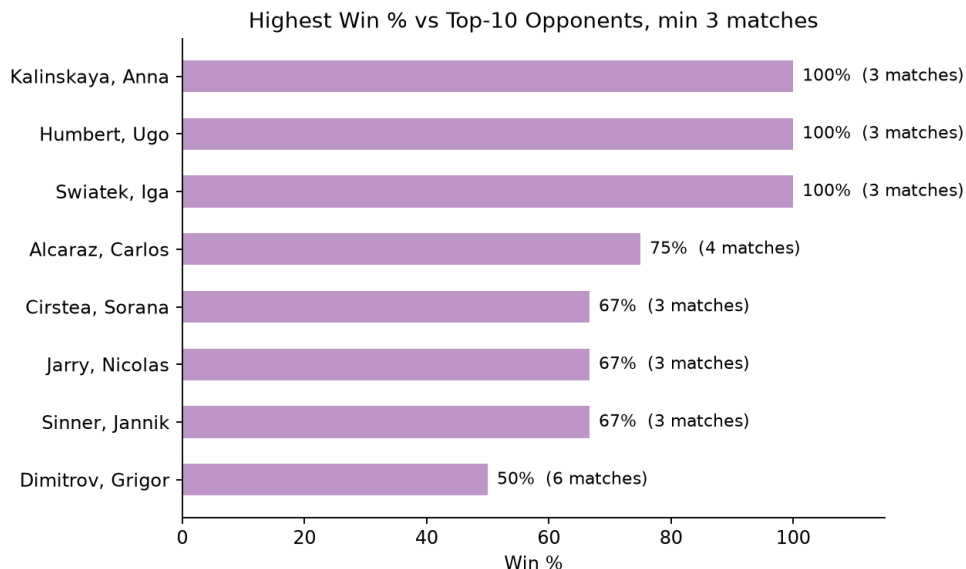
Question 16

Which player has the highest winning percentage against top 10 ranked opponents?

The three players at the top of the table have each played only three matches. In sports analysis, three matches constitute a small sample size; consequently, the performances of Alcaraz (75% win rate over 4 matches) and Dimitrov (50% over 6 matches) are highly significant in terms of statistical consistency.

The code utilizes the ``current_rank`` column. This means the opponent's **current** ranking is used as the criterion, rather than the ranking they held at the time the match actually took place.

As a result, this analysis is not reliable.



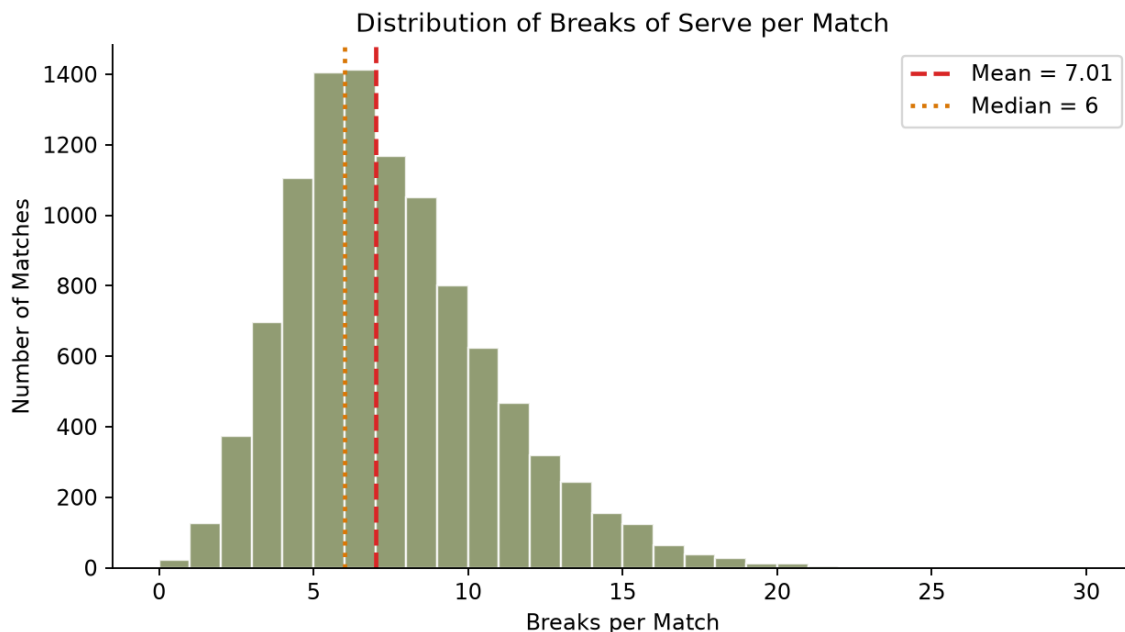
Question 17

What is the average number of breaks of serve per match?

The mean number of breaks per match is 7.01, and the median is 6; the narrow gap between the mean and median indicates a relatively symmetrical distribution, albeit with a slight right skew.

This right-hand tail is caused by 1.6% of the data points falling into the outlier range. We chose not to remove them, as they represent genuine occurrences—likely resulting from long matches characterized by poor serving quality or tight competition between two strong returners.

(There were 36 matches recorded with only 1 or 2 games; validation against the home/away team score data confirmed that these were indeed short matches rather than cases of incomplete data, so they were not excluded.)



bonus / self-designed questions

Optional Question 1

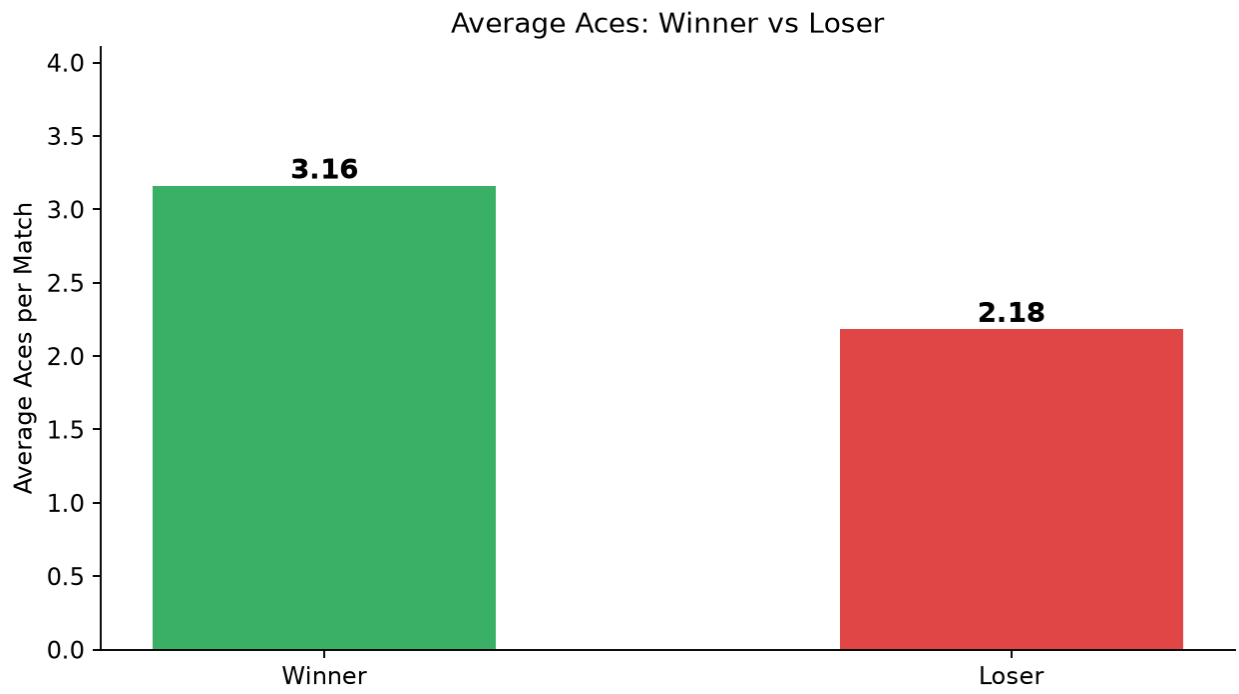
Does the winner usually play more aces?

The average ace count for winners is 3.16, which is approximately 45% higher than the 2.18 average for losers.

However, this pattern does not hold true for every match: winners recorded more aces in only 49.9% of matches, while ace counts were tied in 23% of cases. In the remaining 27%, the loser actually hit more aces but still lost the match; consequently, the number of aces alone is not a definitive predictor of the outcome.

In short, while winners do average more aces, this represents a general statistical advantage rather than an absolute rule for every match.

(Matches where the winner was unknown i.e., the ``winner_code`` was missing were excluded from the analysis to avoid making assumptions about the outcome.)



Optional Question 2

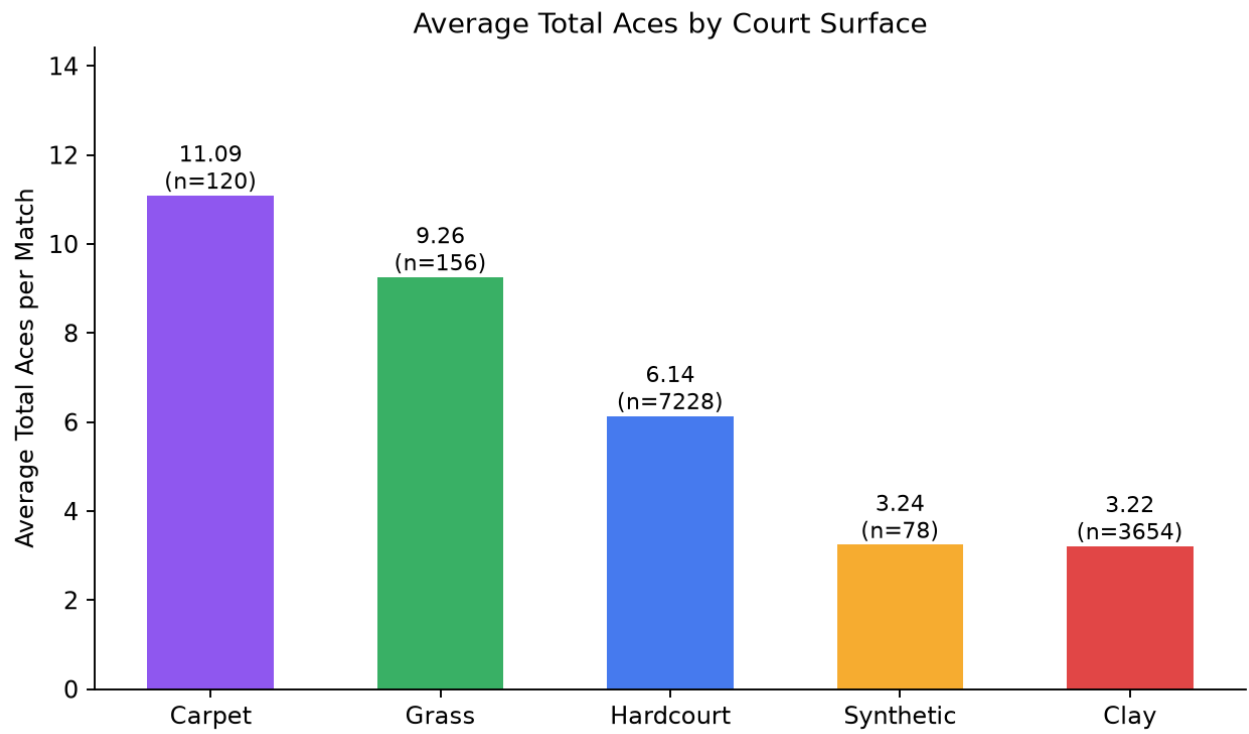
Does the court surface affect the number of aces?

Fast courts produce significantly more aces than slow courts a difference of nearly 3.5 times between the highest and lowest rates.

This pattern aligns perfectly with the physics of tennis: on clay, the ball moves more slowly and bounces higher, making service returns easier; conversely, carpet and grass surfaces maintain ball speed, thereby increasing the likelihood of aces.

Consequently, the court surface plays a major role (grass and carpet yield the highest number of aces, while synthetic and clay yield the lowest).

(To ensure statistical significance, raw and fragmented surface categories—such as green clay and indoor red clay—were consolidated into five main groups: Hardcourt, Clay, Grass, Carpet, and Synthetic.)



Optional Question 3

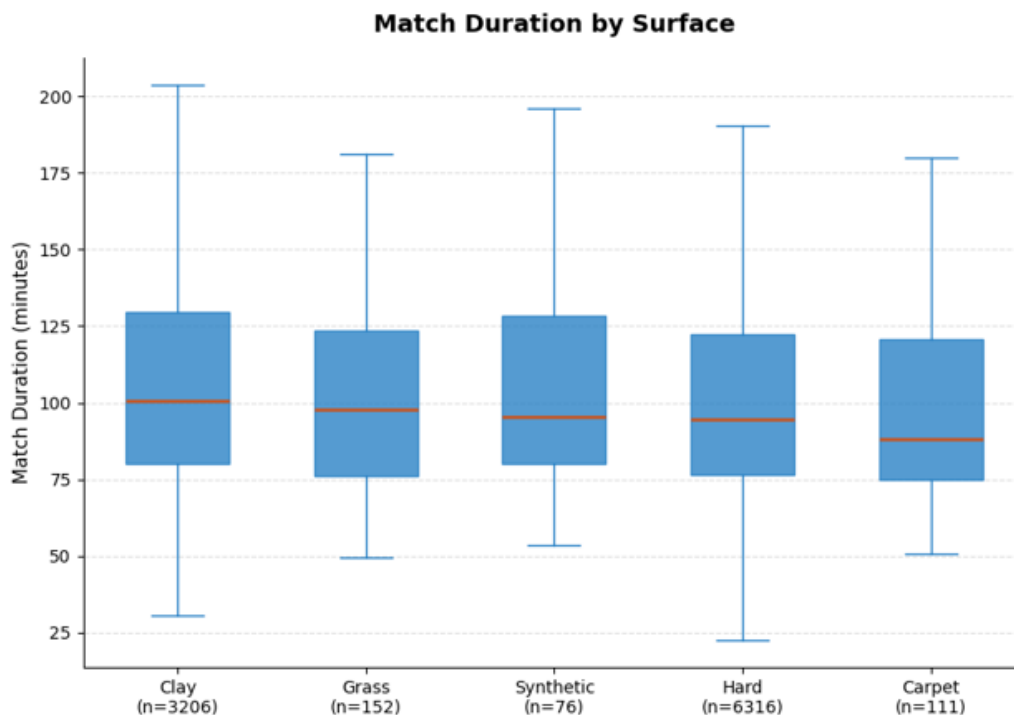
Does the duration of the match depend on the level of play?

Finding

Median minutes under the same duration pipeline: Clay 100.8 (n=3,206), Grass 98.0 (n=152), Synthetic 95.4 (n=76), Hard 94.6 (n=6,316), Carpet 88.2 (n=111). Outliers hidden on the boxplot.

Meaning

The only large-sample contrast is Clay vs Hard (~6 minutes at the median). Grass / Synthetic / Carpet medians rest on much smaller samples and are less stable.



Optional Question 4

Does player performance change with a shift in court surface particularly between clay and hard courts?

Finding

Among 96 players with ≥ 5 matches on both surfaces, mean within-player difference (Clay – Hard) is **+6.8 pp** (55 better on Clay, 36 on Hard, 5 tied). Paired t-test: **$t = 2.94$, $p = 0.004$** .

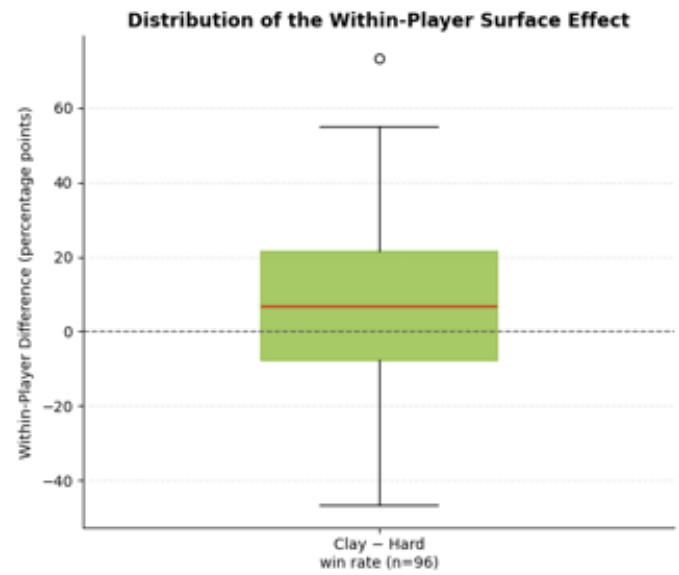
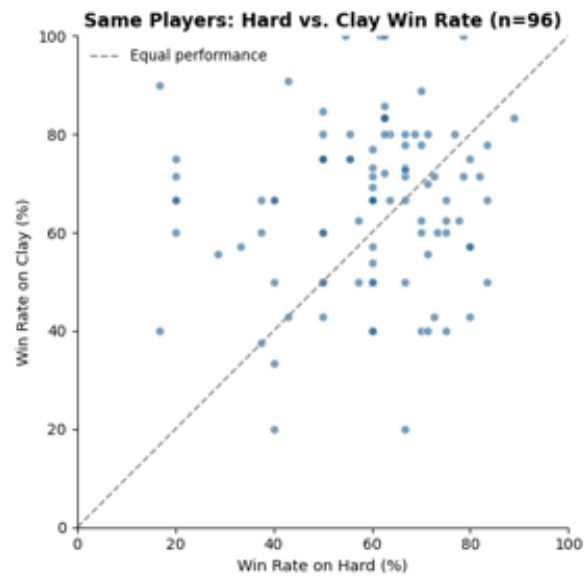
What the test answers

Each player is their own control. The test asks whether the average of (Clay rate – Hard rate) differs from zero across these 96 players.

What it does not answer

p is not effect size. +6.8 pp is modest; individual differences spread widely, so the mean does not describe most players. The shift is not attributed to surface physics, schedule, or fatigue — only measured.

Result depends on the ≥ 5 -match threshold and on the mix of events in the dataset. Dependence across a player's matches is not modelled; the t-test is a first-pass average test, not a full causal model.



Optional Question 5

Is the fan vote share systematically biased toward the "home" label, or does it just reflect genuine quality differences?

Results

Overall average home vote share: 52.25% ($t=8.52$, $p<0.0001$ vs 50%) looks like a small but statistically significant home bias at first glance.

BUT after controlling for actual ranking:

- Home = underdog (worse ranked): 32.01% vote share ($n=4,427$)
- Home = favorite (better ranked): 69.52% vote share ($n=5,206$)

Both are strongly and significantly different from 50% in the direction predicted by genuine quality, not by the "home" label.

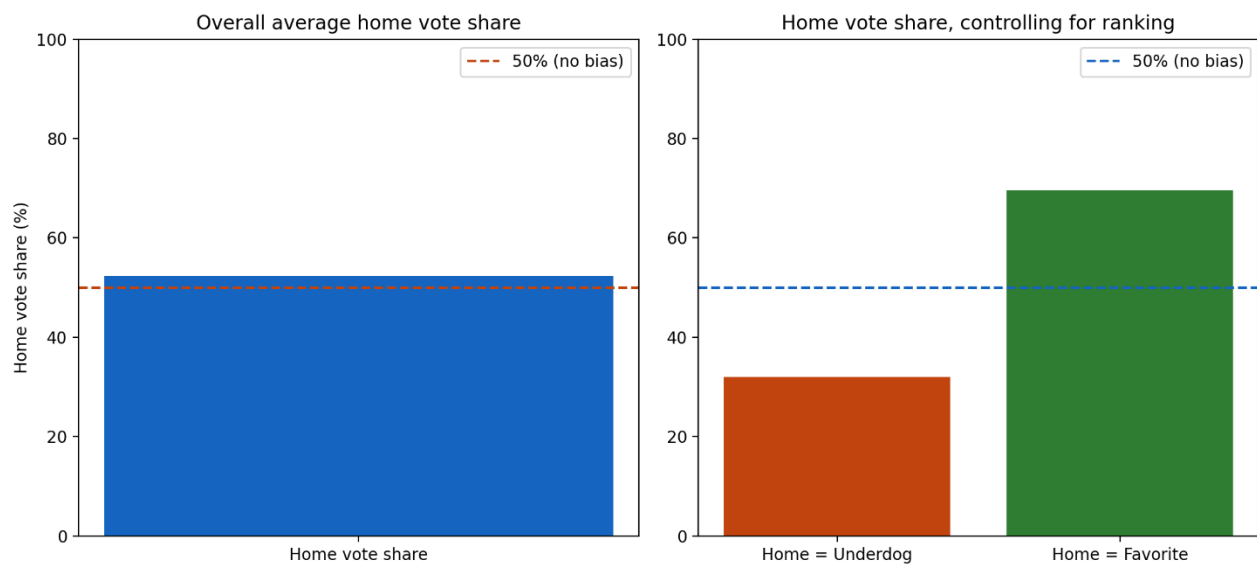
Interpretation

This is a Simpson's-paradox-like result. The overall 52.25% is NOT evidence of a labeling bias -- it is simply the weighted average of two very different, quality-driven behaviors: fans correctly favor whichever player is actually better ranked, regardless of the home/away label. When home happens to be the weaker player, fans correctly shift their votes toward away (only 32% for home). When home happens to be the stronger player, fans correctly favor home (69.52%).

The reason the raw overall average sits slightly above 50% is simply that, in this dataset, matches where home is the better-ranked player (n=5,206) slightly outnumber matches where home is the underdog (n=4,427) -- a sample composition effect, not a psychological bias toward the word "home".

Conclusion

The "labeling bias" hypothesis is REJECTED. Fan vote share tracks genuine player quality (ranking) closely, both when home is favored and when home is the underdog. This is evidence that fan votes carry real predictive information, not blind favoritism toward a display position.



Optional Question 6

Comparing fan votes and the betting market which predicts the actual match winner more accurately?

Results

Comparison	Fan vote accuracy	Betting market accuracy
A complete, readily available example (sets of different competitions)	68.3% (n=13,475 matches)	67.6% (n=9,477 matches)
Fair head-to-head comparison (the same 9,381 matches, accounting for both sources)	68.6%	67.7%

Interpretation

Both prediction sources performed far above the 50% random-guessing baseline (~18 percentage points higher), showing that both fan opinion and betting markets carry real predictive information about match outcomes.

The gap between the two is very small (0.9 percentage points in the fair head-to-head comparison) -- likely within statistical noise given the sample size, rather than a clearly meaningful advantage for either side. In other words, fans and the betting market performed roughly equally well at predicting winners in this dataset.

This is a somewhat counter-intuitive finding

betting markets are often assumed to outperform crowd opinion because real money is at stake. A plausible explanation here is that

this dataset is dominated by lower-tier tournaments (ITF/Challenger, not top ATP/WTa events), where neither the betting market nor fans likely have access to much deeper analysis than the other -- narrowing the usual gap between "informed" market pricing and casual fan sentiment.

Methodology note

matches where votes/odds were tied, missing, or had no market data were excluded from the relevant accuracy calculation, so both percentages reflect only the subset of matches where a clear prediction could be made.

